Q.a. Explain thermal instability and thermal runaway. Define stability factor and find stability factor for any one configuration of a transistor.

Ans. We know that,

$$I_C = \beta I_B + (\beta+1) I_{CBO}$$

→ As temperature increases, I_{CBO} also increase and then I_C increases. Also as I_C increases, ~~I_{CBO}~~ ~~also~~ junction temperature also increases and the I_{CBO} also increases more.

This process goes on in a cumulative way. As a result large collector current flow. This process is called thermal instability and thermal runaway.

Thermal runway can be achieved by a suitable design. To avoid thermal run way following condition must be ~~sat~~ satisfied.

i.e. $$\frac{dP_C}{dT_j} < \frac{dP_D}{dT_j}$$

where $P_C \rightarrow$ heat generated at collector junction and ~~P_D is maxim~~

$P_D \rightarrow$ maximum heat generated.

<u>Stability factor:</u>

Stability factor is defined as the ratio of change in collector current to change in reverse leakage current in CB junction. ~~is called~~

i.e. stability factor, $$S = \frac{\Delta I_C}{\Delta I_{CBO}}$$

Now,

$$S = \frac{\Delta I_C}{\Delta I_{CBO}} = \frac{dI_C}{dI_{CBO}} \quad \text{--- (1)}$$

Also,

$$I_C = \beta I_B + (\beta+1) I_{CBO} \quad \text{--- (2)}$$

From (1) and (2),

$$S = \frac{d(\beta I_B + (\beta+1) I_{CBO})}{dI_{CBO}}$$

$$\Rightarrow S = \frac{d\beta I_B}{dI_{CBO}} + (\beta+1)\frac{dI_{CBO}}{dI_{CBO}}$$

$$\Rightarrow S = \beta \frac{dI_B}{dI_{CBO}} + (\beta+1)$$

Now, differentiating (2) w.r.t. I_C, we get:-

$$\frac{dI_C}{dI_C} = \frac{\beta dI_B}{dI_C} + (\beta+1)\frac{dI_{CBO}}{dI_C}$$

$$\Rightarrow 1 = \beta \frac{dI_B}{dI_C} + (\beta+1)\frac{dI_{CBO}}{dI_C}$$

$$\Rightarrow 1 = \beta . \frac{dI_B}{dI_C} + (\beta+1)\frac{dI_{CBO}}{dI_C}$$

$$\Rightarrow 1 - \beta \frac{dI_B}{dI_C} = (\beta+1) . \frac{1}{S} \quad (\because \text{from (1)})$$

$$\Rightarrow S = \frac{\beta+1}{1 - \beta \frac{dI_B}{dI_C}}$$

$$\therefore S = \frac{dI_C}{dI_{CBO}} = \frac{\beta+1}{1 - \beta \frac{dI_B}{dI_C}}$$

This is required expression for stability factor.

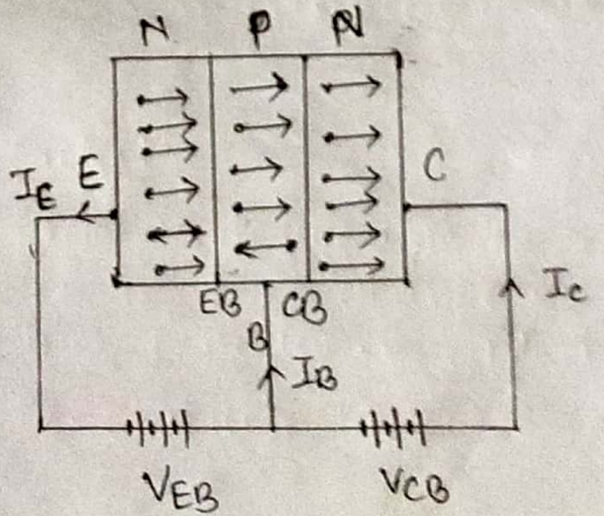
2 a. What is bipolar junction transistor, and what are their applications. Explain the operation of NPN transistor with its current direction.

→ Bipolar junction transistor (BJT) is a three terminal, three regions and two junction device. The three terminals of a transistor are: collector (C), Emitter (E) and Base (B). In this transistor, current is conducted due to both electrons and holes, so it is named as Bipolar junction transistor (BJT).

Applications of BJTs:

i) BJT is used in logic circuits.
ii) BJT is used as an oscillator.
iii) It is used as an amplifier.
iv) It is used in electronics switch.
v) It is used as detector or demodulator.
vi) It is also used as modulator.
vii) It is used in clipping circuits for wave shaping.

Current flow mechanism in NPN Transistor:



The NPN transistor is shown in the above figure in wh emitter base junction is forward biased and collect biase junction is reversed biased.

The forward biased causes the electrons in the N-type to flow towards base. This constitutes the emitter current I_E.

~~As these electrons flow towards base~~

As these electrons flow through p-type base, they tends to combine with holes. As the base is lightly doped and very thin, only few electrons (less than 5% combines with holes to constitude base current I_B.

The remaining electrons (more than 95%) crossover the collector region to constitude collector current I_C.

In this way, almost the entire emitter current flows in the collector circuit. The emitter current is the sum of base ~~of~~ current and collector current.

i.e. $I_E = I_B + I_C$

CamScanner

9.a. Explain the output characteristics of common base transistor. Also derive the relationship between α, β and γ.

Sol: The output characteristics of common base transistor is the curve between output collector current I_C and collector base voltage (output) V_{CB} at constant input emitter current I_E.

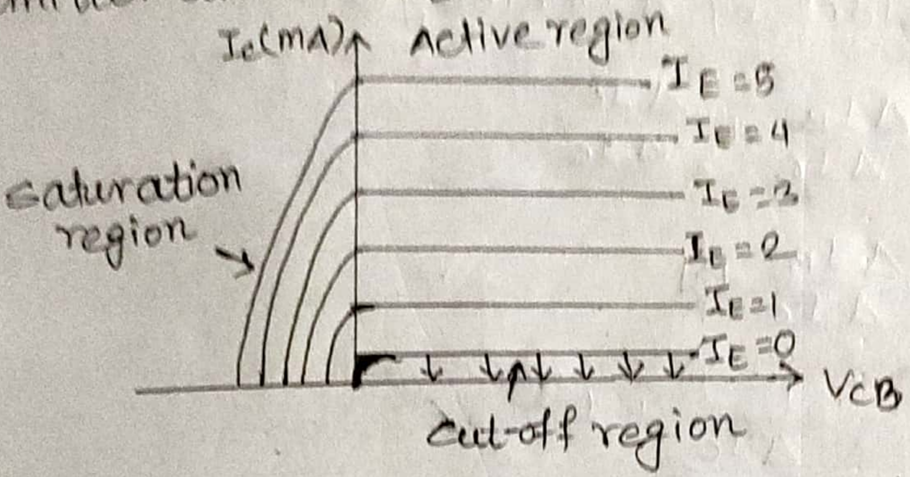


fig. o/p characteristics of CB transistor

→ I_C is nearly equal to I_E in active region. Thus output resistance is very high.

$$\text{i.e. } r_o = \frac{\Delta V_{CB}}{\Delta I_C} \text{ at constant } I_E$$

→ $I_C \approx 0$ in cut-off region i.e. only a small ~~leakage~~ leakage saturation current flows.

→ For a small negative voltage of V_{CB}, EB and CB junctions remain in forward biased condition ~~and~~ in saturation region.

Relationship between α, β and γ.

We know that,

$$\alpha = \frac{\Delta I_C}{\Delta I_E} \;;\; \beta = \frac{\Delta I_C}{\Delta I_B} \;;\; \gamma = \frac{\Delta I_E}{\Delta I_B}$$

Also, $\Delta I_E = \Delta I_B + \Delta I_C$ —— ①

Dividing both sides by ΔI_C, we get,

$$\frac{\Delta I_E}{\Delta I_C} = \frac{\Delta I_B}{\Delta I_C} + \frac{\Delta I_C}{\Delta I_C}$$

$$\Rightarrow \frac{\Delta I_E}{\Delta I_C} = \frac{\Delta I_B}{\Delta I_C} + 1$$

$$\Rightarrow \frac{1}{\frac{\Delta I_C}{\Delta I_E}} = \frac{1}{\frac{\Delta I_C}{\Delta I_B}} + 1$$

$$\Rightarrow \frac{1}{\alpha} = \frac{1}{\beta} + 1$$

$$\Rightarrow \frac{1}{\alpha} = \frac{1+\beta}{\beta}$$

$$\therefore \alpha = \frac{\beta}{1+\beta} \quad \text{—— ②}$$

Also,

$$\frac{1}{\alpha} = \frac{1}{\beta} + 1$$

$$\Rightarrow \frac{1}{\alpha} - 1 = \frac{1}{\beta}$$

$$\Rightarrow \frac{1-\alpha}{\alpha} = \frac{1}{\beta}$$

$$\therefore \beta = \frac{\alpha}{1-\alpha} \quad \text{—— ③}$$

Again, $\gamma = \frac{\Delta I_E}{\Delta I_B} = \frac{\Delta I_E}{\Delta I_C} \times \frac{\Delta I_C}{\Delta I_B}$

$$\gamma = \frac{1}{\alpha} \cdot \beta$$

$$\therefore \gamma = \frac{\beta}{\alpha} \quad \text{——— (4)}$$

And,

$$\gamma = \frac{1}{\alpha} \cdot \frac{\alpha}{1-\alpha} = \frac{1}{1-\alpha} \quad (\because \text{using (3)})$$

$$\therefore \gamma = \frac{1}{1-\alpha} \quad \text{——— (5)}$$

Again,

$$\gamma = \frac{\beta}{\frac{\beta}{1+\beta}} = 1+\beta \quad (\because \text{using (2)})$$

$$\therefore \gamma = 1+\beta \quad \text{——— (6)}$$

Eqn. (2), (3), (4), (5) and (6) gives the relationship between α, β and γ.

2.a. Explain BJT as an amplifier and as a switch with required waveform.

Sol

①. BJT as a switch:

To use BJT as as switch, the input voltage V_{in} is supplied to the base through R_B and output voltage V_o is taken from the collector through R_C.

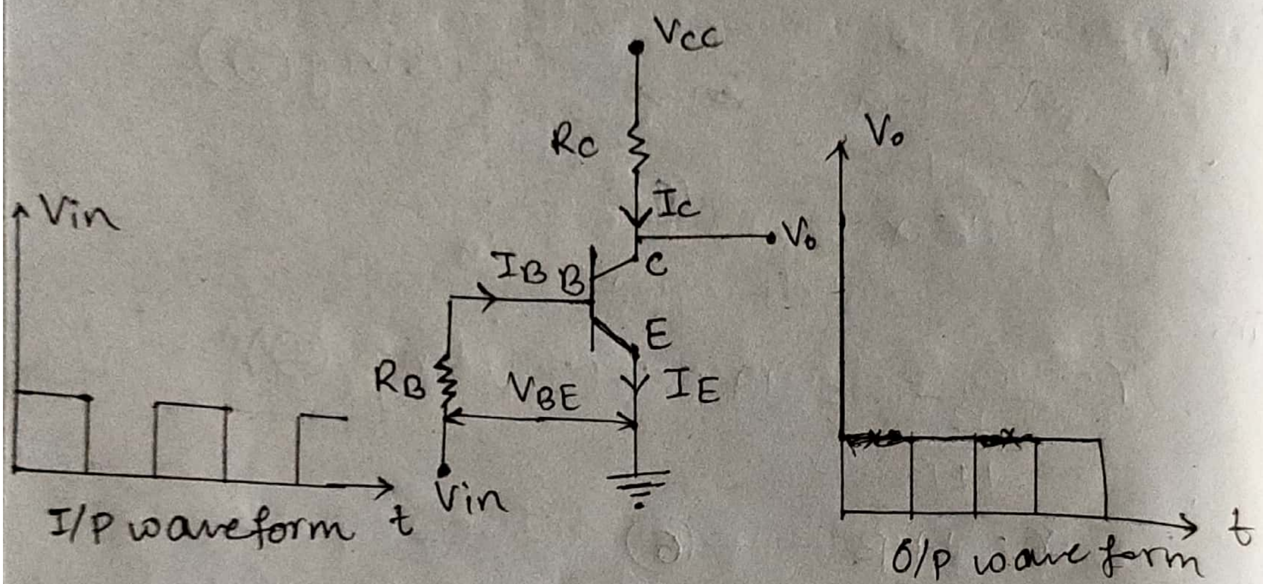


fig- BJT as a switch.

The cut off and saturation mode of operation are used to operate BJT as a switch.

1. Operation in cut-off mode:

→ For normal operation of BJT in active mode, V_{BE} must be greater than V_{th} (0.7 V).

→ If V_{BE} is less than 0.7 V, the base emitter junction will be reverse biased and no current flows through transistor.

→ For $V_{BE} < 0.7V$, $I_B = 0 \Rightarrow I_L = I_E = 0$ $\therefore V_o = V_{cc}$

→ Since all currents are zero, the BJT is in OFF state i.e switch OFF in cut-off mode.

- **Operation in saturation mode:**

→ When V_{BE} is greater than 0.7 V, the base-emitter junction will be forward biased and conduct currents I_C and I_B to ground through emitter.

→ The output voltage nearly equal to zero.

BJT as an Amplifier:

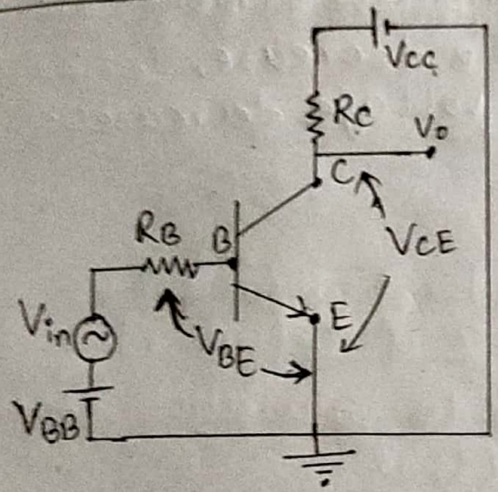


fig. circuit diagram

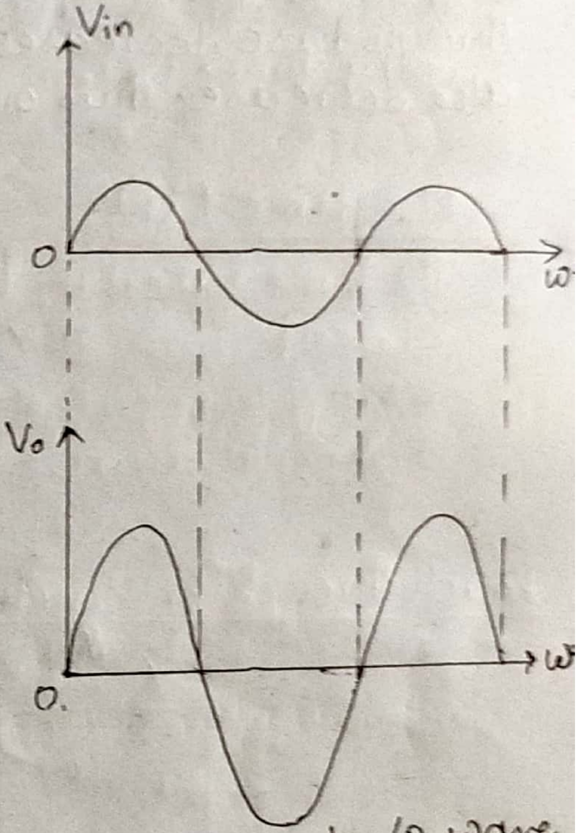


fig. I/P and o/P wave form

→ To use BJT as an amplifier, it must be operated in active mode.

→ Consider a common emitter transistor ~~amplifier~~ circuit as shown in figure has DC power supply V_{BB} and V_{CC} are connected in such a way that they are always forward-biased emitter base junction and reverse-biased collector base junction despite of source voltage V_s.

operation:

→ For positive half cycle of source signal, the base becomes more positive due to this more current flow through base increases. As I_B increases, $I_C = \beta I_B$ also increases. Thus output voltage increases.

→ For negative half cycle of source signal, the base becomes less positive due to this ~~less~~ current flows through base decreases. As I_B decrease, $I_C = \beta I_B$ also decrease. Thus output voltage decreases.

4.a. Why do we need small signal low frequency model of BJT? Describe it.

sol The V-I characteristics curve of BJT is non-linear. The analysis of non-linear device is complex. Thus to simplify the analysis of BJT, its operation is restricted to the linear V-I characteristics around Q-point i.e. in the active region. This approximation is only possible with small input signals. With small input signals, transistor can be replaced with small signal linear model. Therefore, we need small signal low frequency model of BJT.

4.a Make comparison between ~~BC~~ CB, CE, CC.

sol

characteristics	CB	CE	CC
Input resistance	very low (20 Ω)	Low (1 kΩ)	very high (500kΩ
Output resistance	very high (1 MΩ)	High (40 kΩ)	Low (50 Ω)
voltage gain	very high	Moderate	Less than 1
current gain	1	Moderate	very high
phase shift between input & output voltages	0° or 360°	180°	0° or 360°
Applications	For high frequency circuit as pre-amplifier.	For audio frequency circuits.	For impedance matching.

2. a. Whare the Delay time and Rise time? Briefly explain the Reach Through or Punch Through effect.

Sol

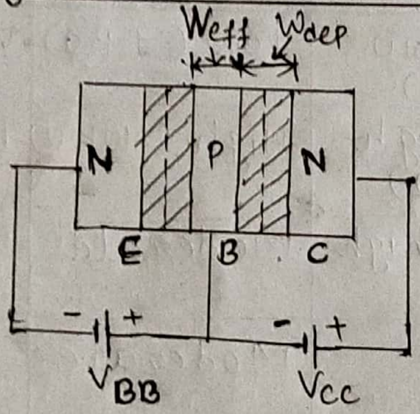
Delay Time (t_d):

The time taken by transistor to increase its current from 0 to 10% of $I_c(sat)$ while moving from cut-off to saturation mode is known as delay time (t_d).

Rise time (t_r):

The time taken by transistor to increase its current from 10% to 90% of $I_c(sat)$ while moving from cut-off mode to saturation mode is known as rise time (t_r).

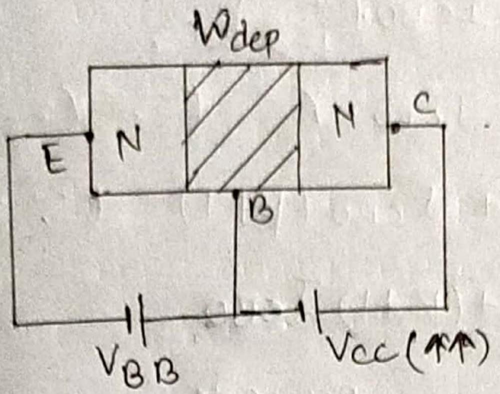
Reach through or Punch through effect:



→ For operation of BJT in active mode, emitter base junction must be forward bias and ~~common~~ collector base junction must be reverse bias.

→ When reverse bias voltage Vcc is increased, the depletion layer ~~of~~ expand over the base region and ~~p~~ base region dissappears. When base region disappears, the collector and emitter ~~[illegible]~~ looks like short circuited.

→ Due to this ~~region~~ reason, high current flow through transistor. This effect is called punch through or Reach through effect.



2.a. show that $I_C = \beta I_B + (\beta+1) I_{CBO}$

Sol: The emitter current for transistor is given by

$$I_E = I_B + I_C \quad \text{——— (1)}$$

The total current flow through collector is given by

$$I_C = \alpha I_E + I_{CBO} \quad \text{——— (2)}$$

where, I_{CBO} → reverse leakage current flow through CB-junction,

$\alpha \rightarrow I_C / I_E$ = current amplification factor

from (1) and (2),

$$I_E = I_B + \alpha I_E + I_{CBO}$$

$$\Rightarrow I_E - I_B = \alpha I_E + I_{CBO}$$

$$\Rightarrow I_C = \alpha I_E + I_{CBO}$$

$$\Rightarrow I_C = \alpha (I_B + I_C) + I_{CBO} \quad (\because \text{from (1)})$$

$$\Rightarrow I_C = \alpha I_B + \alpha I_C + I_{CBO}$$

$$\Rightarrow I_C - \alpha I_C = \alpha I_B + I_{CBO}$$

$$\Rightarrow I_C (1-\alpha) = \alpha I_B + I_{CBO}$$

$$\Rightarrow I_C = \frac{\alpha I_B}{1-\alpha} + \frac{I_{CBO}}{1-\alpha}$$

$$\Rightarrow I_C = \frac{\alpha}{1-\alpha} \cdot I_B + \frac{I_{CBO}}{1-\alpha}$$

$$\Rightarrow I_C = \beta I_B + \frac{I_{CBO}}{1-\alpha} \quad \left(\because \beta = \frac{\alpha}{1-\alpha}\right)$$

$$\Rightarrow I_C = \beta I_B + I_{CBO} \cdot \left(\frac{1}{1-\alpha}\right)$$

$$\Rightarrow I_C = \beta I_B + I_{CBO} \cdot \gamma \quad \left(\because \gamma = \frac{1}{1-\alpha}\right)$$

$$\Rightarrow I_C = \beta I_B + I_{CBO}(\beta+1) \quad (\because \gamma = \beta+1)$$

$$\therefore I_C = \beta I_B + (\beta+1) I_{CBO}$$

proved.

3.a. Define rectification. Explain that the effeciency of full wave rectifier is double of the half wave rectifier.

Sol The process of converting sinusoidal ac signal into a pulsating dc signal is called rectification.

Efficiency is defined as the ratio of dc output power to ac output power.

i.e. $\eta = \frac{P_{dc}\,(O/P)}{P_{ac}\,(I/P)}$

where, $P_{ac} = I_{rms}^2 (R_L + r_D)$, r_D = diode resistance

$P_{dc} = I_{dc}^2 R_L$

Now,

$$\eta = \frac{P_{dc}}{P_{ac}} = \frac{I_{dc}^2 R_L}{I_{rms}^2 (R_L + r_D)}$$

For maximum efficiency, η_{max} ; $r_D \approx 0$. So

$$\eta_{max} = \frac{I_{dc}^2 R_L}{I_{rms}^2 R_L} = \frac{I_{dc}^2}{I_{rms}^2} \quad \text{---} \; (1)$$

For ~~full~~ half wave rectifier,

$$I_{dc} = \frac{I_m}{\pi} \quad \text{and} \quad I_{rms} = \frac{I_m}{2}$$

From (1),

$$\eta_{max(H)} = \frac{(I_m/\pi)^2}{(I_m/2)^2} = \left(\frac{2}{\pi}\right)^2 = 0.46$$

For ~~half~~ full wave rectifier,

$$I_{dc} = \frac{2I_m}{\pi} \quad \text{and} \quad I_{rms} = \frac{I_m}{\sqrt{2}}$$

from (1),

$$\eta_{max(F)} = \frac{\left(\frac{2I_m}{\pi}\right)^2}{(I_m/\sqrt{2})^2} = \left(\frac{2\sqrt{2}}{\pi}\right)^2 = 2 \times 0.46$$

$$\therefore \eta_{max(F)} = 2 \times \eta_{max(H)}$$

This shows that effeciency of full wave rectifier is double that of half wave rectifier.

Q.a. Explain the working principle of Half wave rectifier.

Sol An electronic device which converts AC signal into pulsating DC signal is known as rectifier.

The process of converting sinusoidal AC signal to a pulsating DC signal is known as rectification.

There are two types of rectifiers:

i> Half wave rectifier

ii> Full Wave Rectifier

Half wave Rectifier:

The rectifier circuits which converts any of half cycle of input ac signal to pulsating dc signal is known as half wave rectifier.

It is consists of step down transfer and a diode connected across a load resistor R_L as shown in figure.

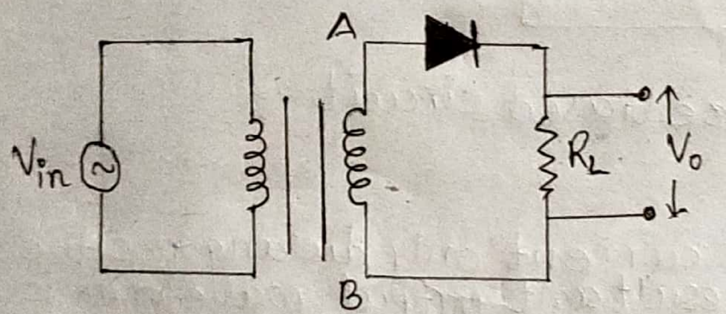


fig. Full Wave Rectifier

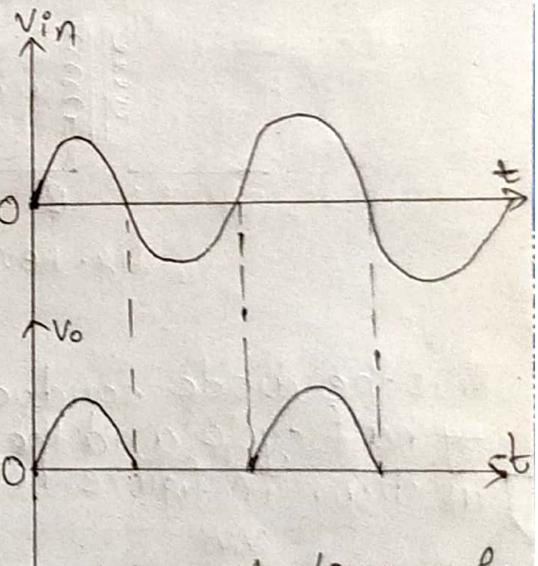


fig. I/P and O/P waveform

Operations:

i> For positive half cycle of input:

⇒ During positive half cycle of input ac signal, the end A of secondary winding of transformer becomes positive and that of end B becomes negative. The diode is forward biased and ideally it behaves as a short circuited so that current I_o flows through load resistor R_L and $V_o = V_{in}$.

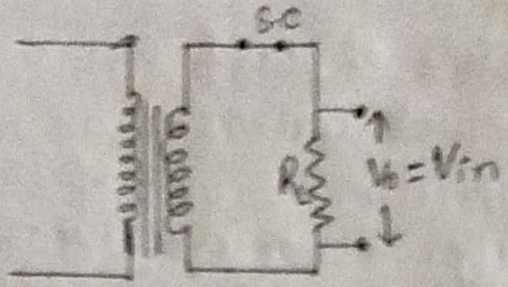


fig. forward biased circuit.

ii) For negative half cycle of input:

⇒ During negative half cycle of input ac signal, the end of ~~seconding~~ secondary winding ~~of~~ of transformer becomes negative and that of end B becomes positive. The diode is reversed biased and ideally it behaves as ~~short~~ open circuited so ~~that~~ the current I_0 doesn't flow through load resistor R_L and $V_0 = 0$.

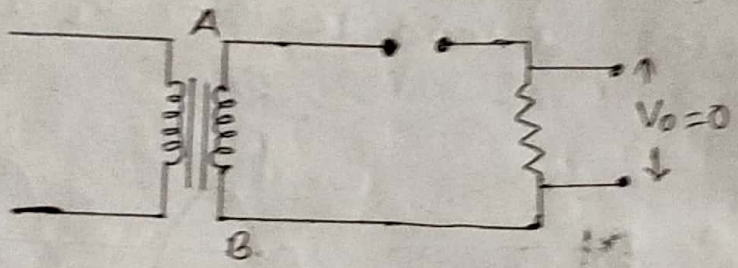


fig. Reverse biased circuit

Thus the diode conducts current only during positive ~~by~~ half cycle and the resultant output waveform is as shown in figure below

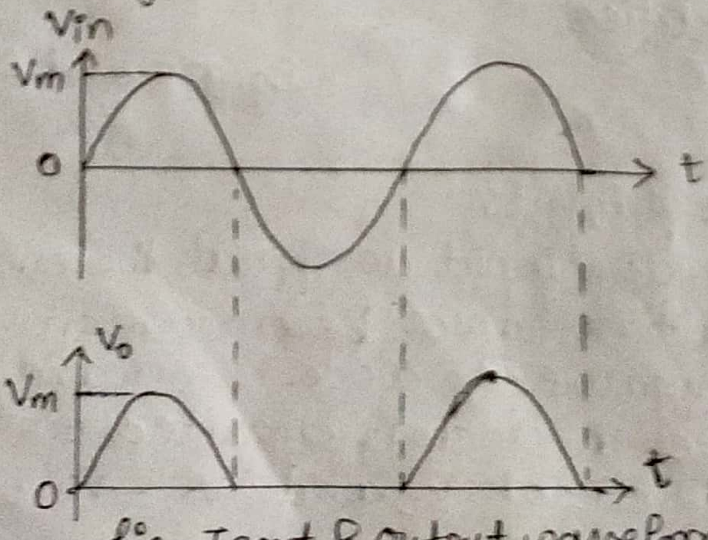


fig. Input & output waveform of HWR.

Full Wave Rectifier (Centre-Tapped)

The rectifier which converts both positive and negative half cycle of input ac signal into a pulsating dc output signal is known as full wave rectifier.

The centre tapped full wave rectifier consists of a centre tapped step down transformer and two ~~diode~~ diodes D_1 and D_2 connected across a load resistor R_L as shown in figure below:

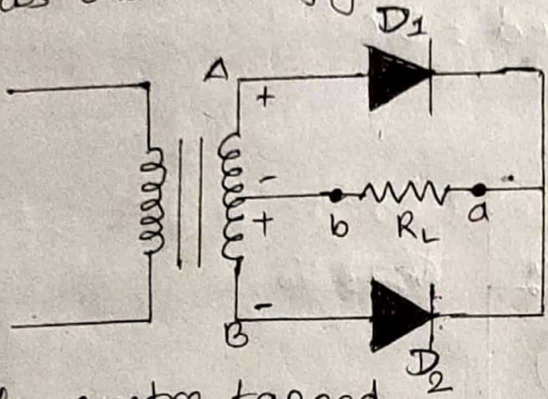


fig. centre-tapped FWR

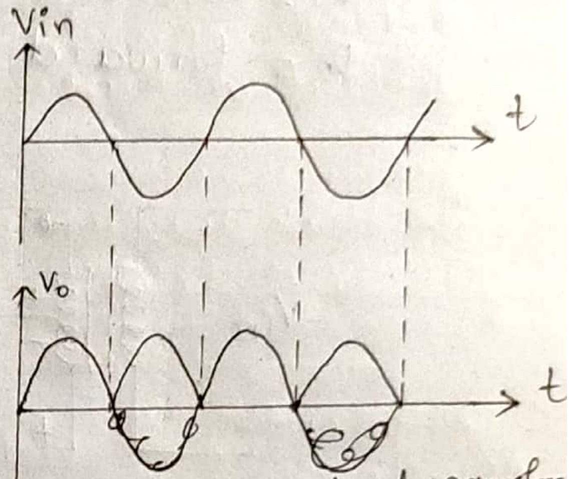
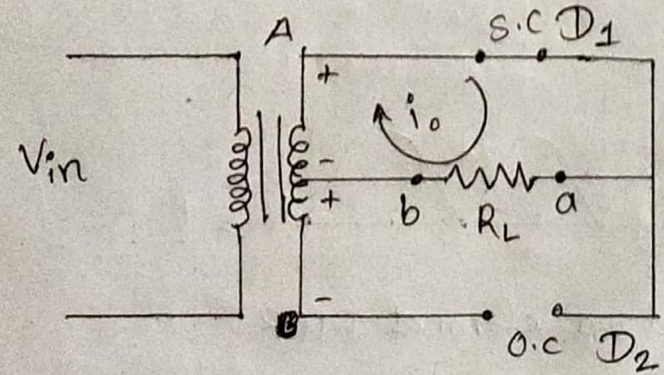


fig. input & output waveform

Operations:

i) ~~During~~ For positive half cycle:

⇒ During positive half cycle of input ac signal, the end A of secondary winding of transformer becomes positive and that of end B becomes negative. Diode D_1 ~~and~~ becomes forward biased and Diode D_2 becomes reverse biased. The forward biased diode D_1 behaves as short circuit and conducts current ~~to~~ through load resistor R_L from end a to end b as shown in figure below:



Here,
D_1 = Forward biased = S.C

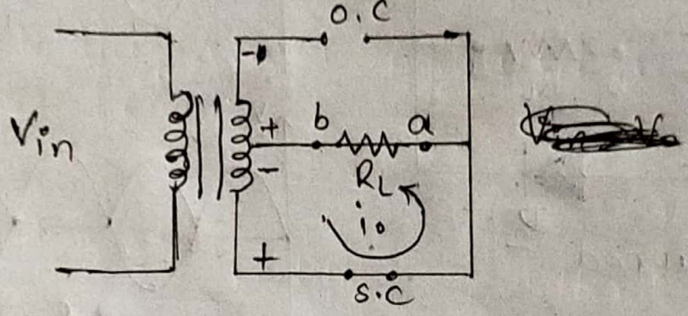
D_2 = Reversed biased = O.C

b) For negative half cycle:

⇒ During negative half cycle of input ac signal, the end A of secondary winding of transformer become negative and that of end B becomes positive. The Diode D_1 is reversed bias and D_2 is forward biased. The forward biased diode conducts current through load resistor R_L from a to b as shown in figure.

Here, D_1 = reversed biased = O.C

D_2 = forward biased = S.C



Thus ~~the dio~~ both the diodes D_1 and D_2 conduct current through load resistor R_L in the same direction. The resultant output waveform as shown in figure.

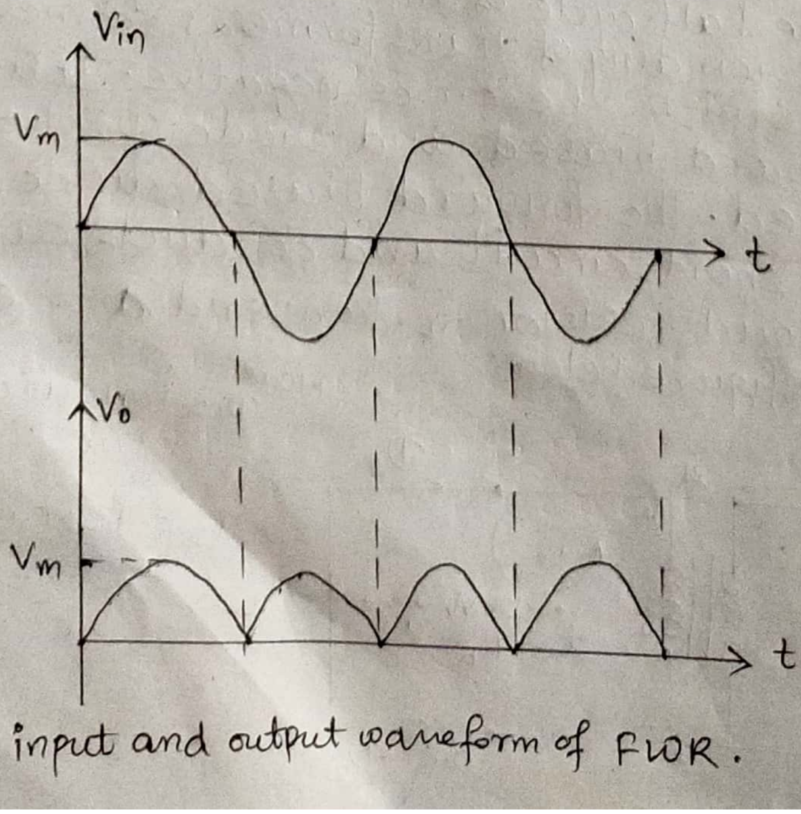


fig. input and output waveform of FWR.

9.a Comparison of all the parameters of bridge centre tapped FWR and Bridge FWR.

Sol

Parameters	Centre Tapped FWR	Bridge FWR
Number of diodes	2	4
Maxium Efficiciency	81.2%	81.2%
Peak Inverse voltage	$2V_m$	V_m
V_{dc}	$\frac{2V_m}{\pi}$	$\frac{2V_m}{\pi}$
Transformer utilization factor	0.693	0.812
Ripple factor	0.48	0.48
form factor	1.11	1.11
Peak factor	$\sqrt{2}$	$\sqrt{2}$
Average current	$I_{dc}/2$	$I_{dc}/2$
output frequency	2f	2f

Sol

3.a. Explain the block diagram of dc regulated power supply with the waveform at each point.

Sol A regulated dc power supply is an electronic circuit which produces a constant output volta[ge] irrespective of the changes in the input voltage or frequecy and independent of the variations in the output load conditions.

The block diagram of regulated dc power supply is shown below:

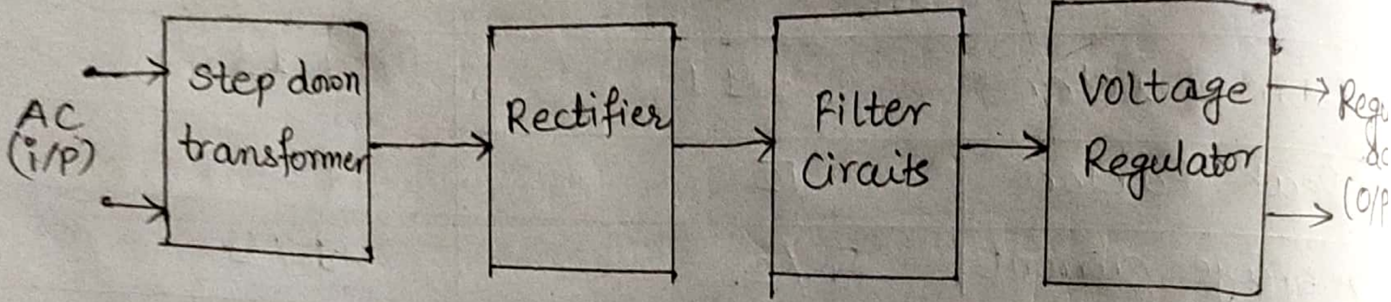


fig. block diagram of regulated dc power supply

There are four blocks in regulated dc power supply which are discussed below:

~~1) Transformer:~~

1) Step down transformer:

⇒ A stepdown transformer is an electrical device that reduces the voltage of an ac power supply. In regulated power supply, it is used to step down the voltage as per requirement.

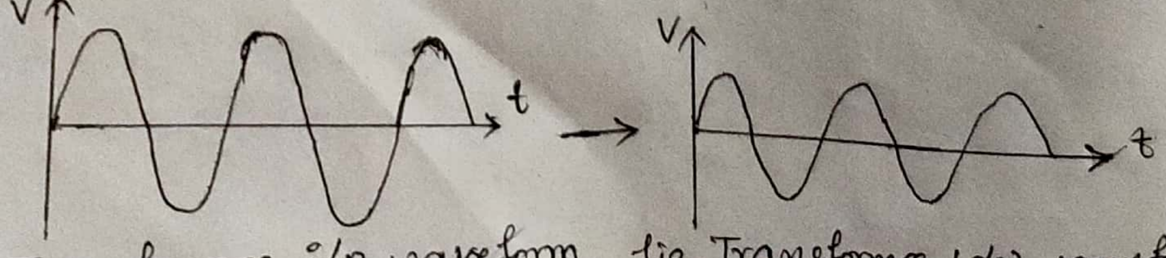


fig. Transformer i/p waveform fig. Transformer (o/p) waveform

2> Rectifier:

An electronic device which converts sinusoidal ac ~~Sieg~~ signal into a pulsating dc signal is called rectifier. In regulated dc power supply, it is used to convert AC signal to DC signal.

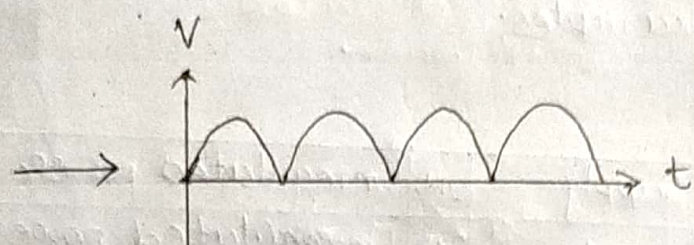


fig. Rectifier output waveform

3> Filter Circuit:

There are some ripples or AC components available in DC ~~power~~ ~~supply~~ signal. To remove those ripples and ~~ai~~ make the signal pure DC, a filter circuit is used.

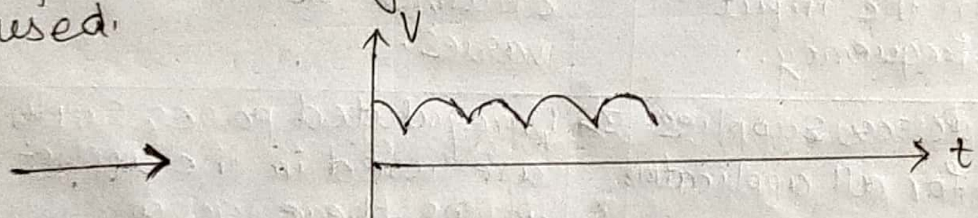


fig: Filter Circuit Output waveform

4> Voltage regulator:

A voltage regulator is the last and most important block of regulated dc power supply because a voltage regulator actually does the regulation

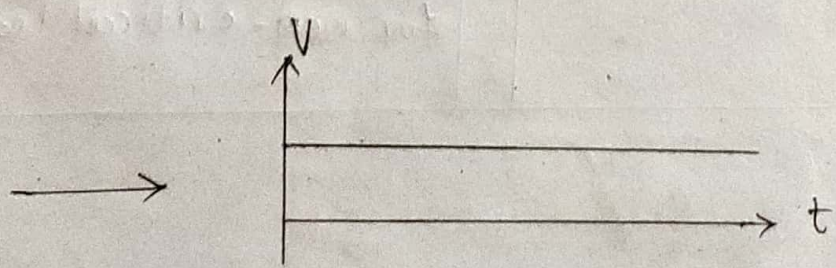


fig. voltage regulator output waveform.

3.a. Sketch the output waveform ... silicon diode.

Sol

3.a. Define regulated and unregulated power supplies with suitable examples:

Sol

S.N	Regulated Power Supply	S.N.	Unregulated power Supply.
1)	Regulated Power supply converts unregulated ac voltage to constant dc voltage.	1)	Unregulated power supply provides a constant output based on the input and load voltage
2)	Regulated Power supply produces a constant output voltage irrespective of the changes in the input voltage or frequency.	2)	Unregulated Power Supply produces an ~~variable~~ output voltage which changes as the load varies.
3)	Regulated Power supplies are used for all applications that requires an exact amount of output voltage.	3)	Unregulated power supplies are used in the applications where good regulation or low ripple is not require.
4)	Regulated Power supplies are used in televisions, mobile chargers, computers, medical and measurement devices.	4)	Unregulated power Supplies are used in LED lamps, relay, solenoides, DC moters and any-thing that is used ~~for non~~ ideal for non-critical loads.

Bridge Full wave rectifier

Bridge full wave rectifier consists of four diodes D1, D2, D3 and D4. These four diodes are arranged in such ~~away~~ a way that only two diodes conduct currents during each half cycle. A load resistor R_L is connected between two terminals as shown in figure.

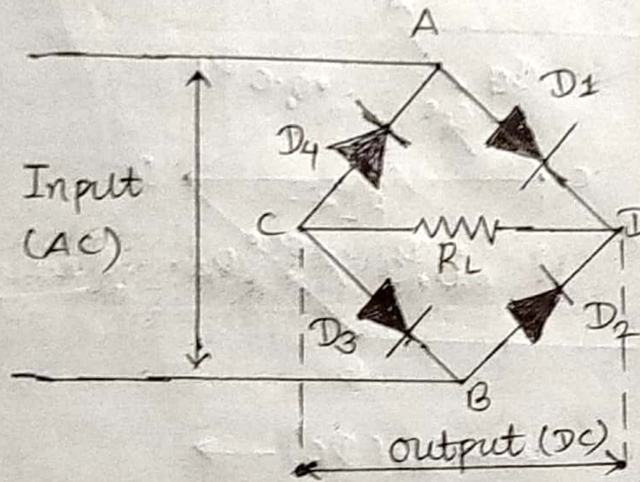


fig. Bridge full wave rectifier

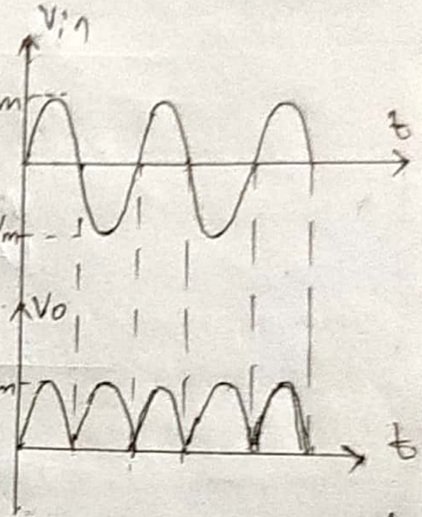
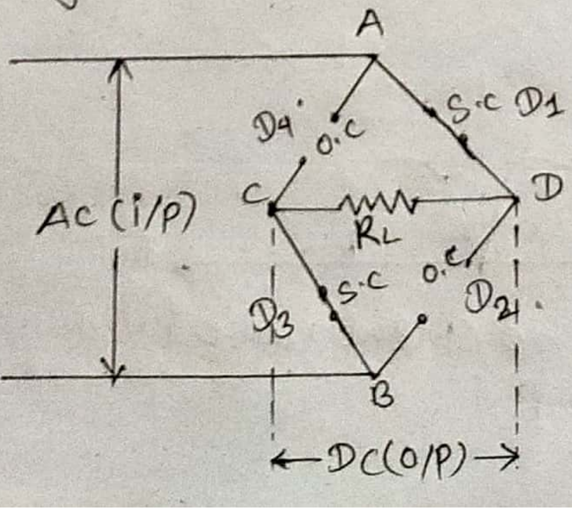


fig. I/P & O/P waveform of Bridge FWR

Operations:

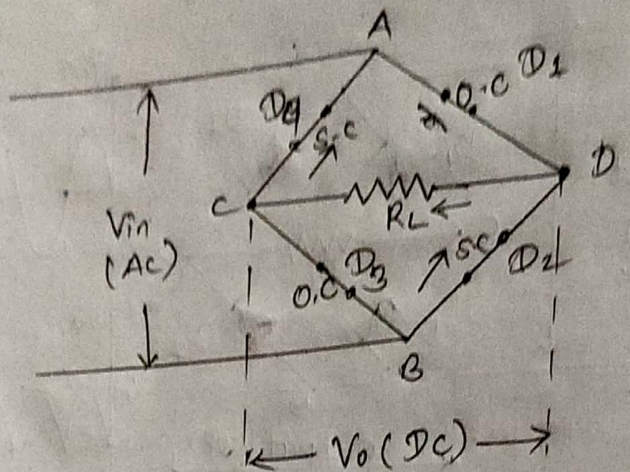
i> For positive half cycle:

⇒ During positive half cycle of input ac signal, the terminal A becomes positive and that of B becomes negative. The diodes D_1 and D_3 becomes forward bias and behave as short circuit while the remaining diodes D_2 and D_4 becomes reversed bias and behaves as open circuited, as shown in figure.



ii) For negative half cycle:

During negative half cycle of input ac signal, the termin[al] A becomes negative while terminal B becomes positive. The diodes D_2 and D_4 becomes forward biased and behave as short circuited while the remaining dio[des] D_1 and D_3 becomes reversed biased and behaves a[s] open circuited as shown in figure.



Thus all diodes D_1, D_2, D_3 & D_4 conducts current throug[h] load resistor R_L and the resultant output wavefo[rm] is show below:

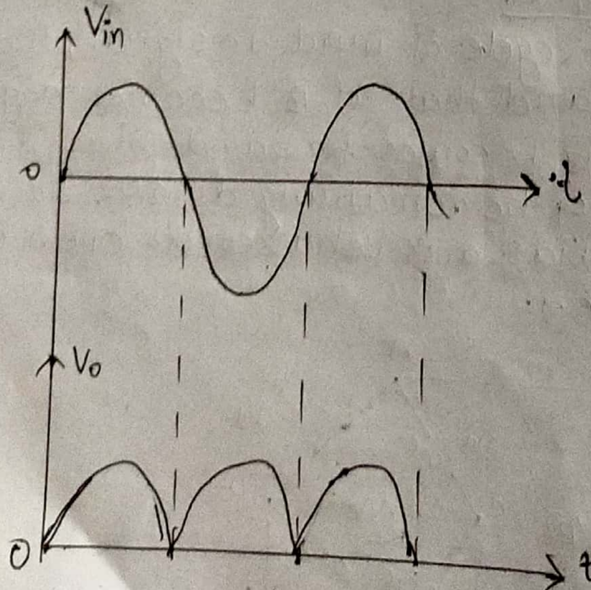


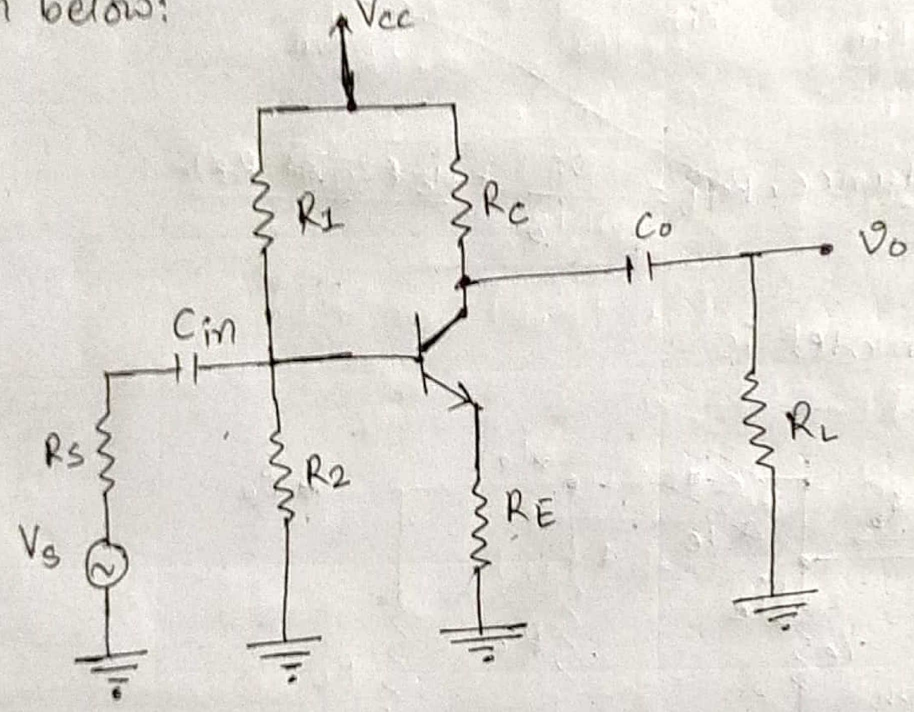
fig. Input and output wave form of FWR

Parameters	Half Wave Rectifier	Full wave rectifier
No. of diodes	1.	→ For center tapped: 2 → For Bridge: 4
Peak Inverse voltage	V_m	→ For center tapped: $2V_m$ → For Bridge : $2V_m$
Maximum efficiency	40.6 %	→ For center tapped: 81.2% → For Bridge : 81.2%
Ripple factor	1.21	→ For center tapped: 0.48 → For Bridge: 0.48
Form factor	1.57	→ For center tapped : 1.11 → For Bridge : 1.11
Peak factor	2	→ For center tapped: 1.41 → For Bridge: 1.41
Transformer Utilization factor	0.2865	→ For center tapped: 0.692 → For Bridge : 0.812
Output frequency	f	→ for center tapped: $2f$ → For Bridge : $2f$

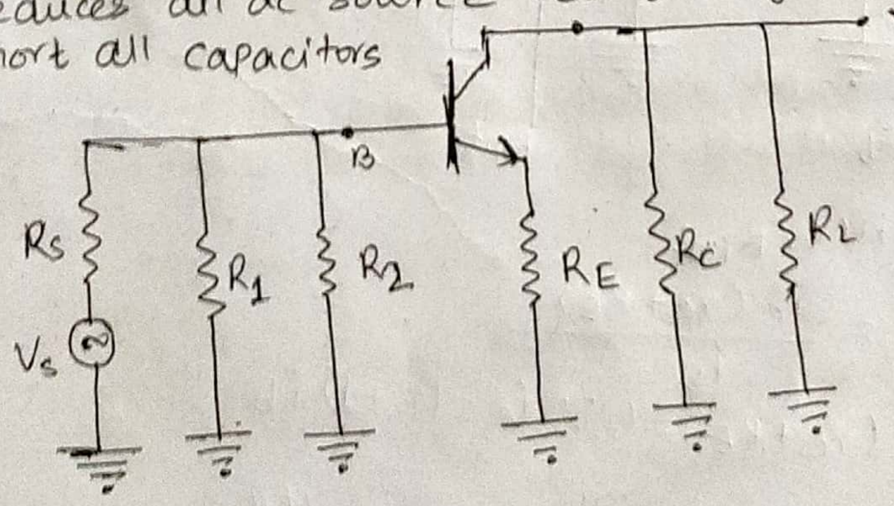
9.a. What are multistage amplifiers? Derive the voltage gain, current gain, input and output impedance and overall voltage gains for common emitter unbypass configuration.

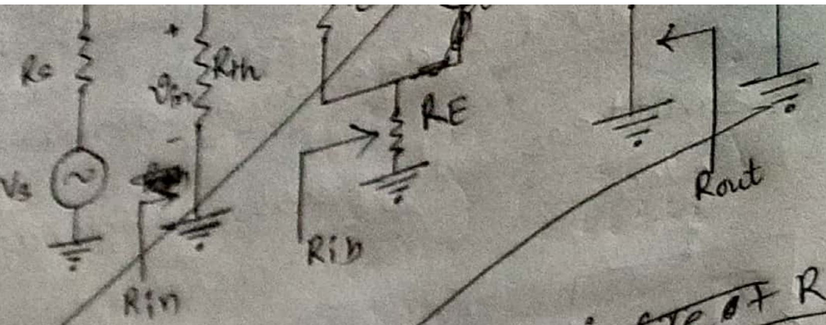
Ans: Multistage amplifiers are those ~~whose~~ where output of first stage is connected with the input of second stage and ~~input~~ output of second stage is connected with input of third stage and so on.

The common emitter unbypass configuration is shown below:



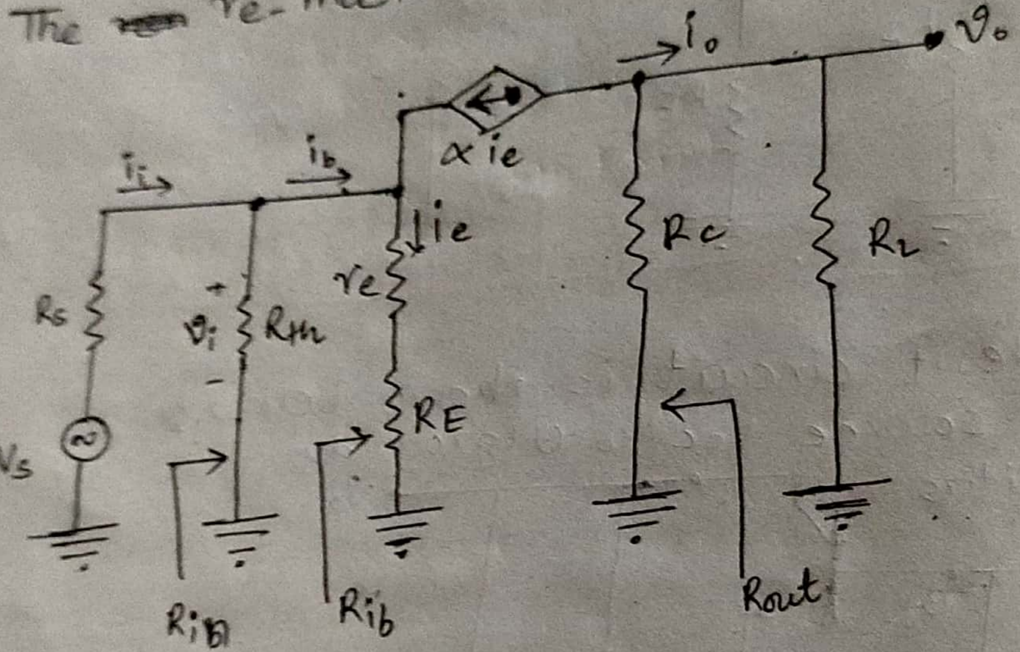
The ac equivalent circuit is shown below:
→ Reduces all dc source Vcc to 0 (ground)
→ short all capacitors





~~i) input impedance, $R_{ib} = \frac{v_i}{i_b} = i_e(r_e + R_E)$~~

The ~~re~~ re-model is



1) input impedance, R_{in}:

$$R_{ib} = \frac{v_i}{i_b} = \frac{i_e(r_e + R_E)}{i_b}$$

$$\therefore R_{ib} = (\beta+1)(r_e + R_E) \quad (\because i_e = (\beta+1)i_b$$

Also, $R_{in} = R_{ib} // R_{th} = (\beta+1)(r_e + r_E) // R_1 // R_2$

$$\therefore R_{in} = (R_1 // R_2) // (\beta+1)(r_e + r_E)$$

2> Output impedance, $R_{out} = R_c$

3> voltage gain, A_v:

since, $A_v = \frac{v_o}{v_i}$ ——(1)

Now,

$$v_o = -\alpha i_e (R_c // R_L)$$

$$= -\alpha (\beta+1) i_b (R_c//R_L) \quad (\because i_e = (\beta+1) i_b)$$

$$= -\beta i_b (R_c//R_L) \quad \left(\because \alpha = \frac{\beta}{\beta+1}\right)$$

$$= \frac{-\beta i_i R_{th}}{(R_{th}+R_{ib})} (R_c//R_L) \quad (\because \text{Applying VDF})$$

$$= \frac{-\beta \left(\frac{v_i}{R_{in}}\right) R_{th}}{(R_{th}+R_{ib})} (R_c//R_L) \quad \left(\because i_i = \frac{v_i}{R_{in}}\right)$$

$$= \frac{-\beta v_i R_{th}}{R_{in}(R_{th}+R_{ib})} (R_c//R_L)$$

$$= \frac{-\beta v_i R_{th}}{(R_{th}//R_{ib})(R_{th}+R_{ib})} (R_c//R_L) \quad (\because R_{in} = R_{th}//R_{ib})$$

$$= \frac{-\beta v_i R_{th}}{\frac{R_{th} \times R_{ib}}{(R_{th}+R_{ib})} \times (R_{th}+R_{ib})} (R_c//R_L)$$

$$\Rightarrow v_o = \frac{-\beta v_i}{R_{ib}} (R_c//R_L)$$

since $R_L \to \infty$. so

$$\therefore v_o = \frac{-\beta v_i}{R_{ib}} (R_c)$$

$$\Rightarrow \frac{v_o}{v_i} = \frac{-\beta R_c}{R_{ib}}$$

$$\therefore A_v = \frac{-\beta R_c}{R_{ib}}$$

4) Current gain, A_i;

we know,

$$i_o = -\alpha i_e \left(\frac{R_c}{R_c + R_L} \right)$$

$$= \cancel{-(\beta+1) i_b}$$

$$\Rightarrow i_o = \frac{-\beta}{(\beta+1)} (\beta+1) i_b \left(\frac{R_c}{R_c + R_L} \right)$$

$$\Rightarrow i_o = -\beta i_b \left(\frac{R_c}{R_c + R_L} \right)$$

$$\Rightarrow i_o = -\beta \, i_i \left(\frac{R_{th}}{R_{th} + R_{ib}} \right) \left(\frac{R_c}{R_c + R_L} \right)$$

$$\Rightarrow \frac{i_o}{i_i} = \frac{-\beta R_{th} R_c}{(R_{th} + R_{ib})(R_c + R_L)}$$

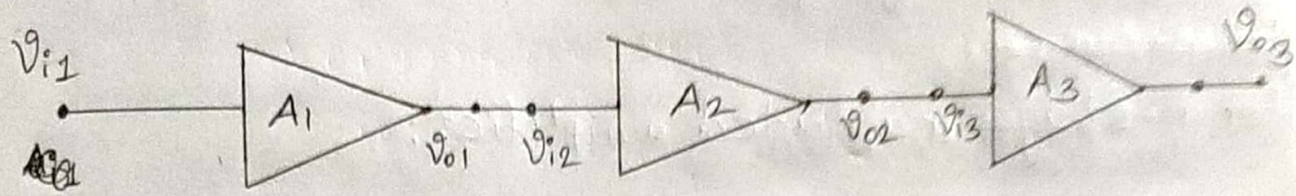
$$\Rightarrow A_i = \frac{-\beta R_{th} R_c}{(R_{th} + R_{ib})(R_c + R_L)}$$

Since $R_L \to 0.80$

$$A_i = \frac{-\beta R_{th}}{(R_{th} + R_{ib})}$$

4.b. Why cascading is necessary? Find the gain of n-stage cascaded amplifier.

Sol: Cascading in amplifier is necessary to get high voltage gain or current gain.



Let us consider three cascaded amplifiers A_1, A_2 and A_3 as shown in figure.

For first stage,

$$A_1 = \frac{v_{o1}}{v_{i1}} \qquad \therefore v_{o1} = A_1 v_{i1} \quad \text{—— (1)}$$

For second stage;

$$A_2 = \frac{v_{o2}}{v_{i2}} = \frac{v_{o2}}{v_{o1}} \qquad (\because v_{i2} = v_{o1})$$

$$\therefore v_{o2} = A_2 A_1 v_{i1} \quad \text{—— (2)}$$

$$\Rightarrow A_2 = \frac{v_{o2}}{A_1 v_{i1}}$$

For third stage;

$$A_3 = \frac{v_{o3}}{v_{i3}} = \frac{v_{o3}}{v_{o2}} \qquad (\because v_{i3} = v_{o2})$$

$$\Rightarrow A_3 = \frac{v_{o3}}{A_2 A_1 v_{i1}} \qquad \cancel{\therefore v_{o3} = A_3 A_2 A_1 v_{i1}} \text{ (3)}$$

$$\Rightarrow v_{o3} = A_3 A_2 A_1 v_{i1}$$

$$\therefore \frac{v_{o3}}{A_{i1}} = A_3 A_2 A_1 \qquad \therefore A = A_3 A_2 A_1 \quad \text{—— (3)}$$

Let ~~A be the total voltage~~ gain for 3-stage ~~cascaded amplifier.~~

$$\cancel{\therefore A = A_n A_{n-1} \cdots A_3 A_2 A_1}$$

~~Hence,~~

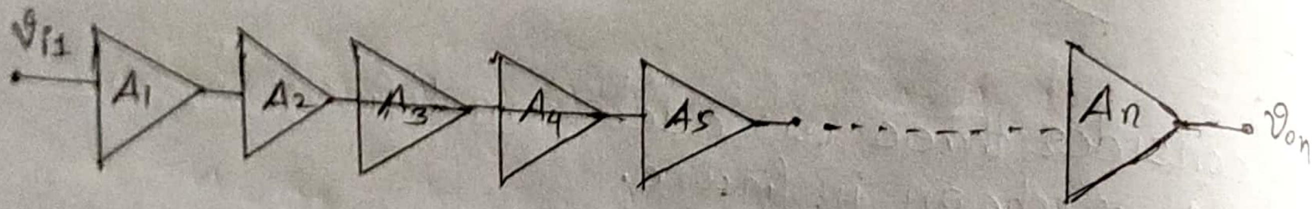
$\therefore$ The total voltage gain A is given by

$$A = A_3 A_2 A_1$$

It concluded that the overall voltage gain of 3 stages ~~multi~~ amplifier is the product of individual stage gain of the amplifier.

~~For n-casca~~

For n-stages cascaded amplifier:



The overall voltage gain of n-stage amplifier is given by

$$A = A_n A_{n-1} \text{-------} A_5 A_4 A_3 A_2 A_1$$

To present, the gain in dB.

$$20 \log A = 20 \log(A_n A_{n-1} \text{-----} A_3 A_2 A_1)$$

$$\Rightarrow 20 \log A = 20 \log A_n + 20 \log(A_{n-1}) + \text{----} + 20 \log(A_3) + 20 \log(A_2) + 20 \log$$

$$\therefore A(dB) = A_n(dB) + A_{n-1}(dB) + \text{-----} + A_3(dB) + A_2(dB) + A_1($$

Therefore, the overall voltage $A(dB)$ in dB is the sum of individual stages gains in dB.

4.b) A 3-stage amplifier has voltage gains of 50, 75 and 100 for the first, second and third stage. Find the overall gain of the amplifiers. Also, express the overall gain in dB. Prove that the total phase shift of n-cascaded amplifier is the sum of the phase shifts introduced by the individual stages.

Sol: Given;

$A_1 = 50$

$A_2 = \cancel{100}\ 75$

$A_3 = 100$

overall voltage gain, $A = ?$

we know,

$A = A_3 A_2 A_1$

$= (100)(75)(50)$

$\therefore A = 375000$ Ans

In dB,

$A(dB) = 20 \log(A)$

$= 20 \log(375000)$

$\therefore A(dB) = 111.48$ dB Ans

Let us consider n-stages ~~amplif~~ cascaded amplifiers with voltage gains $A_n, A_{n-1}, \ldots, A_4, A_3, A_2, A_1$ as shown in figure below.

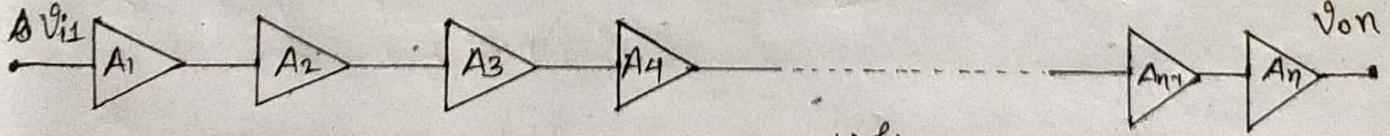


fig. n-stages cascaded amplifier.

For first stage,

$A_1 = \dfrac{V_{o1}}{V_{i1}}$

$\therefore V_{o1} = A_1 V_{i1}$

$\therefore A_1 = A\angle\theta_1$

~~Phase shift, = $A_1 \angle\theta_1$~~

For second stage,

~~Phase shift~~

$$A_2 = A_2 \angle \theta_2$$

For third stage: A

$$A_3 = A_3 \angle \theta_3$$

Similarly,

For n-stages,

$$A_n = A_n \angle \theta_n$$

Since $A = A_n \cancel{A_2 A_3} A_{n-1} \text{------} A_3 A_2 A_1$

$$\Rightarrow A \angle \theta = A_n \angle \theta_n \cdot A_{n-1} \angle \theta_{n-1} \text{------} A_3 \angle \theta_3 A_2 \angle \theta_2 A_1 \angle \theta_1$$

$$\therefore A = A_n A_{n-1} \text{------} A_3 A_2 A_1 \text{ and}$$

$$\theta = \theta_n + \theta_{n-1} \cancel{+\theta} \text{----} + \theta_3 + \theta_2 + \theta_1$$

Thus it showed that the total phase shift of n-stage cascaded amplifier is the sum of phase shift introduced by the individual stages.

Q.6. What are multistage amplifiers. Which configuration of amplifier is chosen in the intermediate stage and why?

Sol Multistage amplifiers are those where the output of first stage ~~are conne~~ is connected with the input of second stage and the output of second stage is connected with the input of third stage and so on.

Common emitter (CE) configuration ~~is used~~ of amplifier is choosen in the intermediate stage while cascading because CE configuration has high current gain and high voltage gain and also moderate input and output resistance.

3.6. Darlington pair amplifier is called super beta ... Derive the expression for it.

sol Darlington pair amplifier is a compound struc... containing two transistors connected in such a... that the current amplified by first transistor is further amplified by second transistor.

A darlington pair amplifier consists of two matched transistors Q_1 and Q_2 with respective current amplification factor β_1 and β_2 as shown in figure.

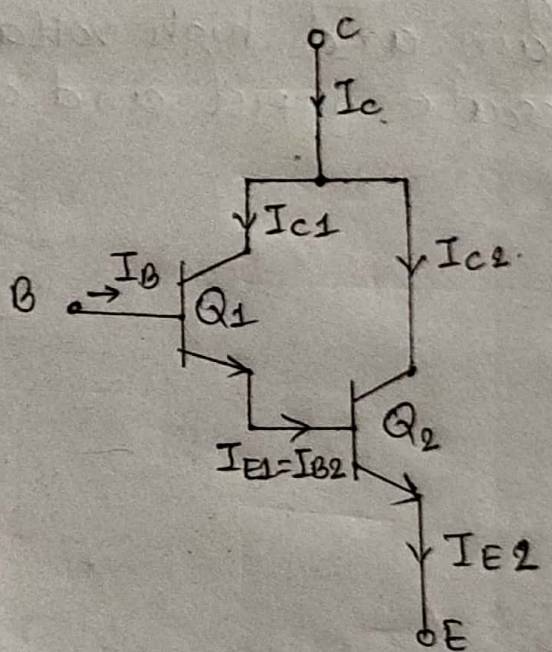


fig. Darlington pair amplifier

From the circuit,

$\therefore \quad I_{C1} = \beta_1 I_{B1}$

$\therefore \quad I_{E1} = (\beta_1 + 1) I_{B1}$

Similaly,

$I_{C2} = \beta_2 I_{B2}$

$= \beta_2 I_{E1} \qquad (\because I_{B2} = I_{E1})$

$\therefore \quad I_{C2} = \beta_2 (\beta_1 + 1) I_{B1}$

CamScanner

Now, $I_C = I_{C1} + I_{C2}$

$= \cancel{(\beta_1+1) I_{B1}}$

$\Rightarrow I_C = \beta_1 I_{B1} + \beta_2(\beta_1+1) I_{B1}$

$\Rightarrow I_C = \beta_1 I_B + (\beta_2 \beta_1 + \beta_2) I_B \quad (\because I_{B1} = I_B)$

$\Rightarrow I_C = \beta_1 I_B + \beta_2 \beta_1 I_B + \beta_2 I_B$

$\Rightarrow I_C = (\beta_1 + \beta_1 \beta_2 + \beta_2) I_B$

$\Rightarrow \dfrac{I_C}{I_B} = \beta_1 + \beta_1 \beta_2 + \beta_2$

Since $\dfrac{I_C}{I_B} = \beta_D$, darling~~ton~~ amplifier gain,

$\therefore \beta_D = \beta_1 + \beta_1 \beta_2 + \beta_2$

~~If $\beta_1 \approx \beta_2 \approx \beta$ then,~~

$\cancel{\beta_D = \beta + \beta^2 + \beta = 2\beta + \beta^2}$

$\Rightarrow \beta_D = \beta_1 + \beta_2 + \beta_1 \beta_2$

Since $\beta_1 \beta_2 >> \beta_1 + \beta_2$. So

$\beta_D = \beta_1 \beta_2$

Since Q_1 and Q_2 are generally matched transistors. So

$\beta_1 = \beta_2 = \beta$

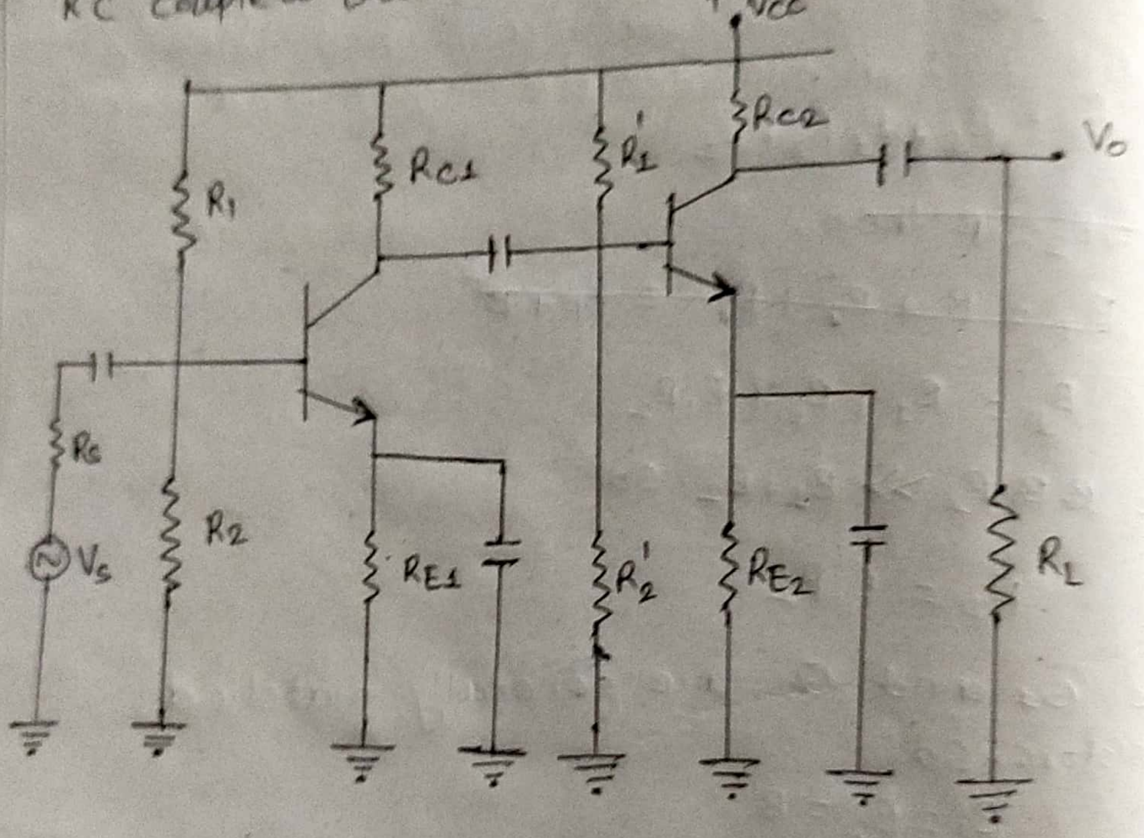
$\therefore \beta_D = \beta . \beta = \beta^2$

Thus Darlington pair amplifier is called superbeta transistor.

4.b. Why common emitter configuration of BJT is u[...] in intermediate stages while cascading. Give reasons. Draw two stage RC coupled transis[tor] amplifier and write its advantages over di[rect] coupling.

Ans Common emitter configuration of BJT is u[...] in intermediates stages while cascading because CE configuration has high voltage gain and current gain and also ~~Input~~ moderate input and output resistance.

RC coupled transistor amplifier:



Advantages of RC coupled amplifier over direct coupling:

i) RC coupled amplifier uses resistors and capacitors which are not expensive so the cost is low.

ii) The circuit is very compact and extremely light.

iii) The frequency response of RC coupled amplifier is excellent.

iv) It offers a constant gain over a wide frequency ba[nd]

Emitter follower

The voltage gain of common collector configuration is given by

$$A_v = \frac{R_E}{r_e + R_E}$$

$$\Rightarrow \frac{V_o}{V_{in}} = \frac{R_E}{r_e + R_E} \quad (\because A_v = V_o/V_{in})$$

Since $R_E >> r_e$ so,

$$\frac{V_o}{V_{in}} = \frac{R_E}{R_E} = 1$$

$$\therefore V_o = V_{in}$$

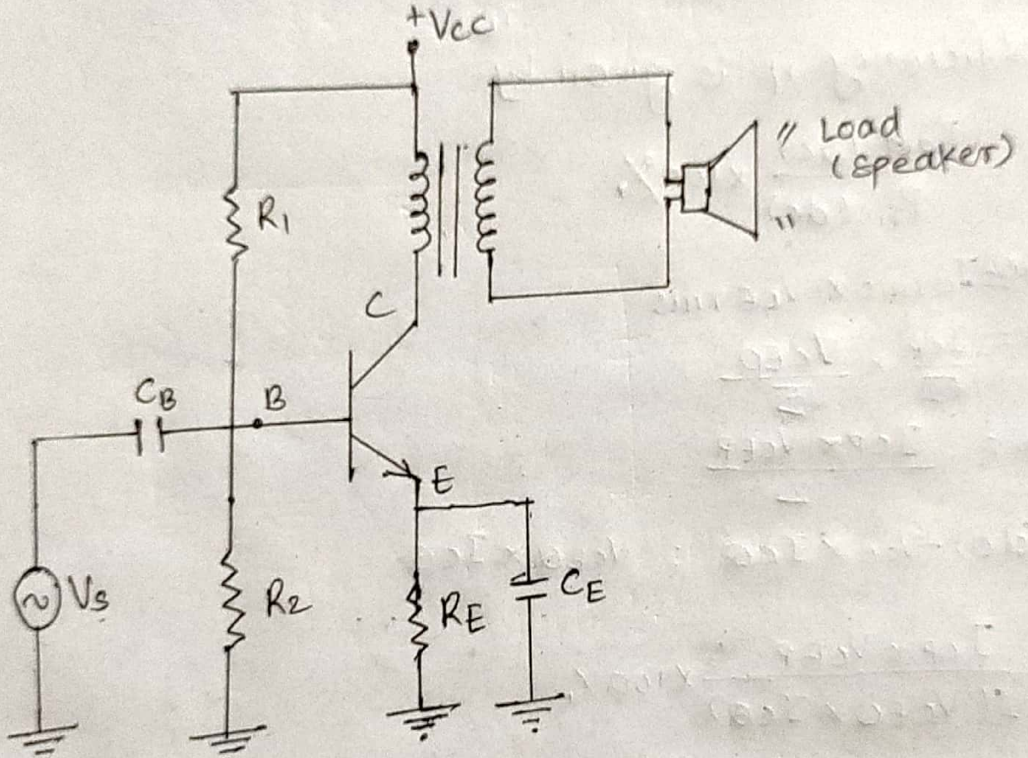
In common collector configuration, input ~~sup~~ V_{in} is supplied to base whereas output V_o is taken from emitter. Here, we get output voltage is ~~equal~~ approximately equal to input voltage. Thus the common collector is also called emitter follower.

CamScanner

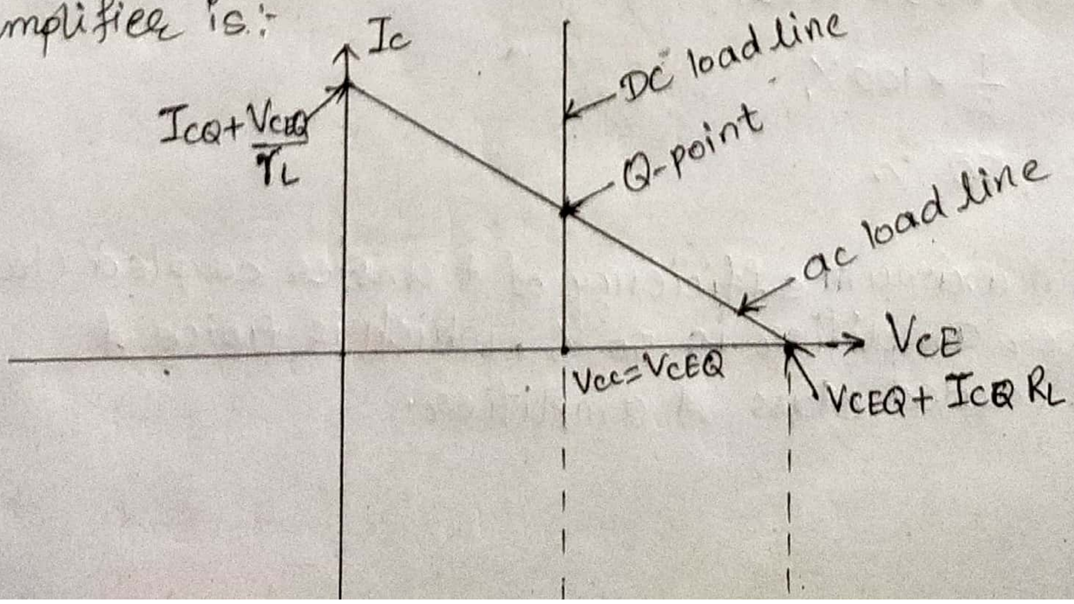
6.b Derive the efficiency for tansformer coupled class A power amplifier with necessary diagram and wave form. List out its draw back.

Sol In a transfoimer coupled class A power amplifier, Transformer is used to couple power amplifier to their loads.

The circuit diagram of transformed coupled class A power amplifier is shown below.



The wave form of transformer coupled class A power amplifier is:-



Here, we assume dc resistance of primary winding is infinitesimally small. The dc load line is almost vertical and then,

$$V_{CC} = V_{CEQ}.$$

The ~~resist~~ ac resistance (r_L) reflected to primary side is given by

$$r_L = \left(\frac{N_P}{N_S}\right)^2 \times R_L$$

where N_P and N_S are no. of turns in primary and secondary windings respectively.

The slope of ac load line is $-\frac{1}{r_L}$

Now, The efficiency η is given by

$$\eta = \frac{P_{o}(ac)}{P_{in}(dc)} \times 100\%$$

where $P_{o}(ac) = I_{C\,rms} \times V_{CE\,rms}$

$$= \frac{I_{CP}}{\sqrt{2}} \times \frac{V_{CEP}}{\sqrt{2}}$$

$$\therefore P_o = \frac{I_{CP} \times V_{CEP}}{2}$$

And $P_{in}(dc) = V_{CC} \times I_{CQ} = V_{CEQ} \times I_{CQ}$

Now,

$$\eta = \frac{I_{CP} \times V_{CEP}}{2(V_{CEQ} \times I_{CQ})} \times 100\%$$

$$= \frac{I_{CQ} \times V_{CEQ}}{2(I_{CQ} \times V_{CEQ})} \times 100\% \quad (\because I_{CQ} = I_{CP} \text{ \& } V_{CEP} = V_{CEQ})$$

$$= \frac{1}{2} \times 100\%$$

$$\therefore \eta = 50\%$$

Hence, the maximum efficiency of transfer coupled class A power amplifier is 50%, which is twice of direct coupled class A amplifier.

5.a. "Large signal amplifier is also called power amplifier." Explain. For a class B amplifier providing a 10 V peak signal to the 20Ω load and a power supply of $V_{cc} = 20$ V. Find the input and output power and efficiency of that amplifier.

Sol^n: The amplifier circuits which handle large input ac signal are called large signal amplifier. The large signal amplifiers are designed to provide a large amount of output ac power so that they can operate the output devices like speaker. Thus, large signal amplifier is also called power amplifier.

For a class B amplifier,

Given;

Peak voltage, $V_{peak} = 10$ V

Load resistor, $R_L = 20\Omega$

DC power supply, $V_{cc} = 20$V

Input power, $P_{in} = ?$

Output power, $P_o = ?$

Efficiency, $\eta = ?$

Now,

Peak Current, $I_p = \frac{V_p}{R_L} = \frac{10}{20} = 0.5$ A

$\therefore$ Input power, $P_{in}(dc) = \frac{2 I_p V_{cc}}{\pi} = \frac{2 \times 0.5 \times 20}{\pi}$

$= 6.366$ W

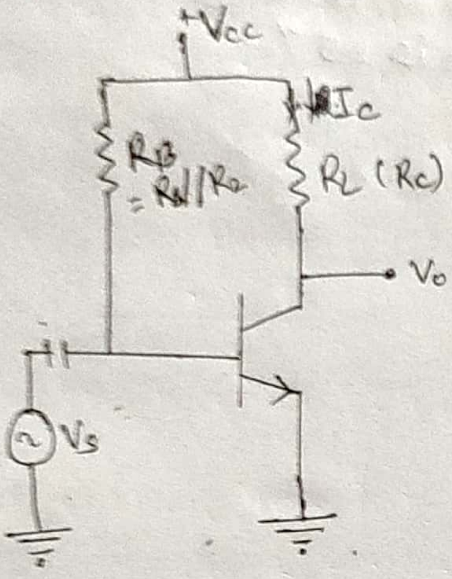
$\therefore$ Output power, $P_o(ac) = \frac{V_p}{\sqrt{2}} \times \frac{I_p}{\sqrt{2}} = \frac{10 \times 0.5}{2} = 2.5$ W

$\therefore$ Efficiency, $\eta = \frac{P_o(ac)}{P_{in}(dc)} = \frac{2.5}{6.366} \times 100\% = 39.27\%$

6

6.a. Show that the efficiency of direct coupled load class A amplifier is 25% with necessary circuit diagram and derivation.

Sol: The direct coupled class A amplifier is also called 'series fed power amplifier' because the load is connected in series with transistor as shown in figure.



The transistor is biased in such a way that it's operating point lies in active region during a complete cycle.

The efficiency η is given by

$$\eta = \frac{P_o(ac)}{P_{in}(dc)} \times 100\%$$

where, $P_o(ac) = I_{C rms} \times V_{CE rms}$

$$= \frac{I_{CP}}{\sqrt{2}} \times \frac{V_{CEP}}{\sqrt{2}}$$

$$\therefore P_o(ac) = \frac{I_{CP} \times V_{CEP}}{2}$$; (I_{CP} = peak collector current, V_{CEP} = peak collector-Emitter voltage)

And,

$$P_{in}(dc) = V_{CC} \times I_{CQ}$$

Now,

$$\eta = \frac{I_{CP} \times V_{CEP}}{2 \times V_{CC} \times I_{CQ}} \times 100\%$$

For maximum efficiency,

$V_{CEP} = \cancel{V_{CC}}\ \frac{V_{CC}}{2}$ and $I_{CP} = I_{CQ}$ and $V_{CEQ} = \frac{V_{CC}}{2}$

$$\therefore \eta = \frac{I_{CQ} \times V_{CC}/2}{2 V_{CC} \times I_{CQ}} \times 100\%$$

$$= \frac{100}{4}\%$$

$$\therefore \eta = 25\%$$

Thus, the maximum possible efficiency of a direct coupled class A amplifier is 25%.

6.d. Draw class B amplifier and derive the expression of its efficiency.

sol: The transistor operation is said to be class B amplifier when the output current varies only during only one half cycle of a sinusoidal wave input.

The ~~cla~~ transistor is biased in such a way that the operating point is set to the cut-off region.

The collector current flows for a half cycle only. so the ~~op~~ output waveform is highly distorted. The waveform is not suitable for audio amp. application.

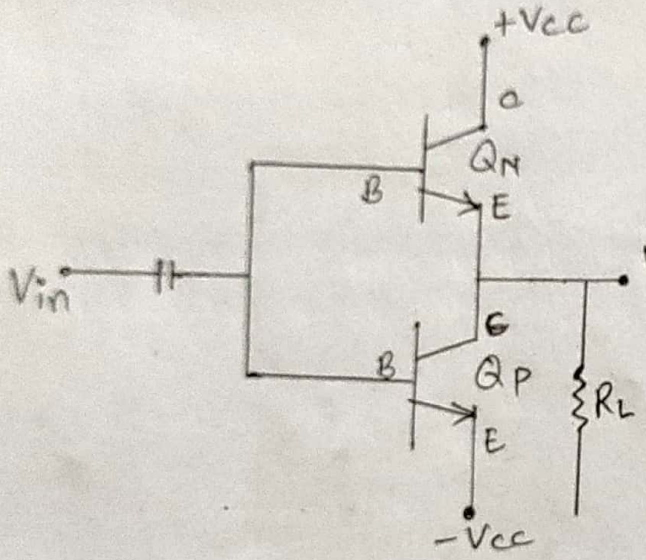
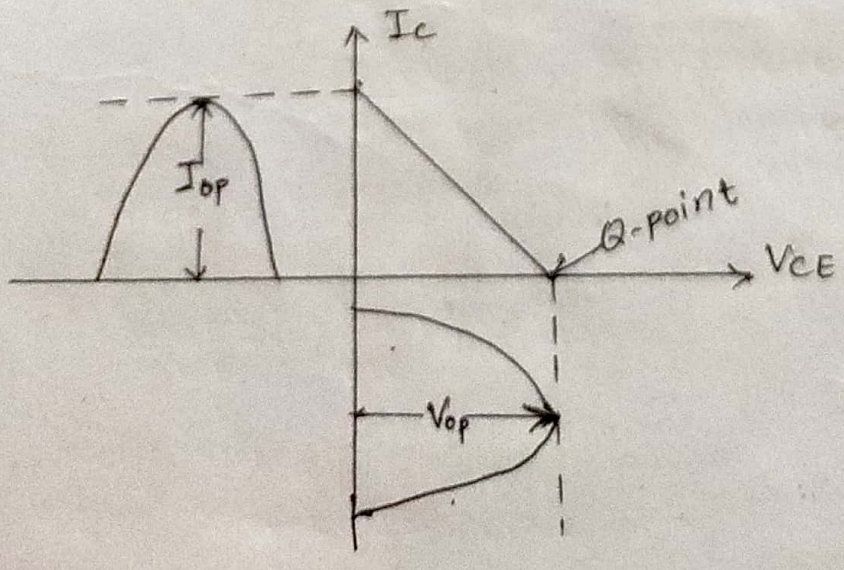


fig. Circuit diagram of class B amplifier.



The efficiency η is given by

$$\eta = \frac{P_{in}(ac)}{P_{in}(dc)} \times 100\%$$

where,

$$P_o(ac) = \frac{I_{Crms} \times V_{CErms}}{2} \quad (\because \text{only one half wave})$$

$$= \left(\frac{I_{CP}}{\sqrt{2}} \times \frac{V_{CEP}}{\sqrt{2}}\right) \times \frac{1}{2}$$

$$\therefore P_o(ac) = \frac{(I_{CP} \times V_{CEP})}{4}$$

And, $P_{in}(dc) = V_{CC} \times I_{dc} = V_{CC} \times \frac{I_{CP}}{\pi}$

Now,

$$\eta = \frac{I_{CP} \times V_{CEP}}{4\left(V_{CC} \times \frac{I_{CP}}{\pi}\right)} \times 100\%$$

$$\therefore \eta = \frac{\pi V_{CEP}}{4 V_{CC}} \times 100\%$$

For maximum efficiency,

$$V_{CEP} = V_{CC}$$

$$\therefore \eta = \frac{\pi V_{CC}}{4 V_{CC}} \times 100\%$$

$$= \frac{100\pi}{4}\%$$

$$\therefore \eta = 78.5\%$$

Hence the maximum efficiency of class B amplifier is 78.5%. Ans

6.a "Efficiency of direct coupled load class A amplifier is half of collector effeciency". verify with necessary derivations.

Sol

i) For ~~maximum~~ direct coupled load class A Amplifier:

$$\text{Efficiency}, \eta = \frac{P_o(ac)}{P_{in}(dc)} \times 100\%$$

where,

$$P_o(ac) = I_{crms} \times V_{CE\,rms}$$

$$= \frac{I_{CP}}{\sqrt{2}} \times \frac{V_{CEP}}{\sqrt{2}}$$

$$\therefore P_o(ac) = \frac{V_{CP} \times V_{CEP}}{2}$$

And

$$P_{in}(dc) = V_{CC} \times I_{CQ}$$

Now,

$$\eta = \frac{I_{CP} \times V_{CEP}}{2(V_{CC} \times I_{CQ})} \times 100\%$$

$$\Rightarrow \eta = \frac{I_{CP} \times V_{CEP}}{2(I_{CQ} \times V_{CC})}$$

For maximum efficiency,

$$V_{CEP} = \frac{V_{CC}}{2} \text{ and } I_{CP} = I_{CQ}$$

$$\therefore \eta = \frac{I_{CP} \times V_{CC}/2}{2(I_{CP} \times V_{CC})} \times 100\%$$

$$= \frac{1}{4} \times 100\%$$

$$\therefore \eta = 25\%$$

ii) For collector efficiency, η_c:

$$\eta_c = \frac{P_o(ac)}{P_{in}(dc)} \times 100\%$$

where, $P_o(ac) = I_{crms} \times V_{CE\,rms} = \frac{I_{CP}}{\sqrt{2}} \times \frac{V_{CEP}}{\sqrt{2}}$

$$\therefore P_o(ac) = \frac{I_{CP} \times V_{CEP}}{2}$$

And $P_{in(dc)} = V_{cc} \times I_{CQ} - I_{CQ}^2 \times R_c$
$= (V_{cc} - I_{CQ} R_c) I_{CQ}$

Now, $$\eta_c = \frac{\cancel{V_{cp} \times V_{ce}} \; I_{cp} \times V_{CEP}}{2(V_{cc} - I_{CQ} \times R_c) I_{CQ}} \times 100\%$$

For maximum ~~pos~~ efficiency,

$$I_{cp} = I_{CQ} = \frac{V_{cc}}{2R_c}$$

$$V_{CEP} = \frac{V_{cc}}{2}$$

$$\therefore \eta_c = \frac{I_{cp} \times V_{CEP}/2}{2(V_{cc} - \frac{V_{cc}}{2R_c} \times R_c) I_{cp}} \times 100\%$$

$$= \frac{V_{cc}}{4(V_{cc} - \frac{V_{cc}}{2})} \times 100\%$$

$$= \frac{V_{cc}}{4 \times \frac{V_{cc}}{2}} \times 100\%$$

$$\therefore \eta_c = 50\%$$

Thus, $\boxed{\eta_c = 2\eta_A}$

Thus, the direct coupled load class A amplifier is half of collector ~~an~~ efficiency.

Cross Over distortion and ~~Sink~~Heat Sink:

In class B push pull amplifier, both transistors remains off until the magnitude of voltage at their base is less than 0.7 V. During that period, there will be no output. As a result output ~~warm~~ waveform will not be in pure sine form but ~~is~~ in distorted form. This type of distortion is known as cross over distortion.

~~Sink~~: Heat sink:

Semiconductor devices are cooled by conduction, radiation and other methodes.

Heat is conducted from a junction through the semi-conductor material and into the surrounding air.

It also radiates surface of the device.

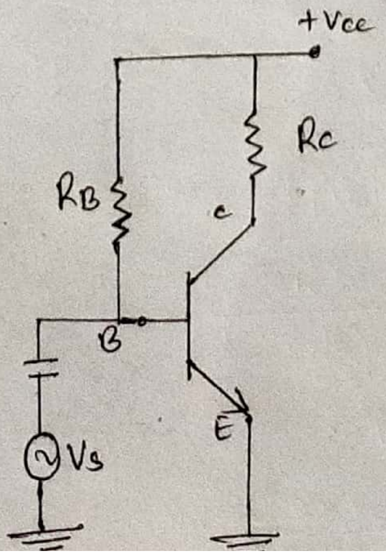
To remove the conduction and rediation of heat from device to the surrounding air, power devices are often equipped with heat sinks.

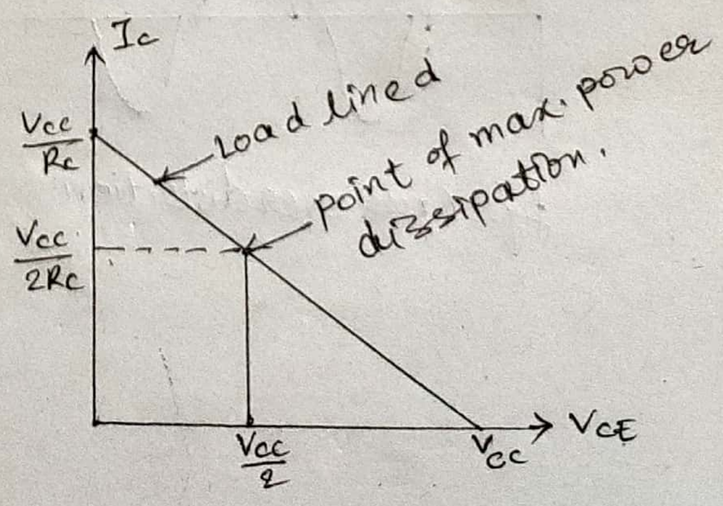
Power dissipation:

For common emitter amplifier, the total power dissipation at the junction is

$$P_d = V_{CB} I_c + V_{BB} I_c$$

$$\cong V_{CE} \cdot I_c$$





The point of maximum power dissipation is given:

$$V_{CE} = \frac{V_{CC}}{2}$$

$$I_C = \frac{V_{CC}}{2R_C}$$

Now, $P_{d\,max} = V_{CC} \cdot I_C$

$$= \frac{V_{CC}}{2} \times \frac{V_{CC}}{2R_C}$$

$$\therefore P_{d\,max} = \frac{V_{CC}^2}{4R_C}$$

Wave form of Cross-Over distortion:

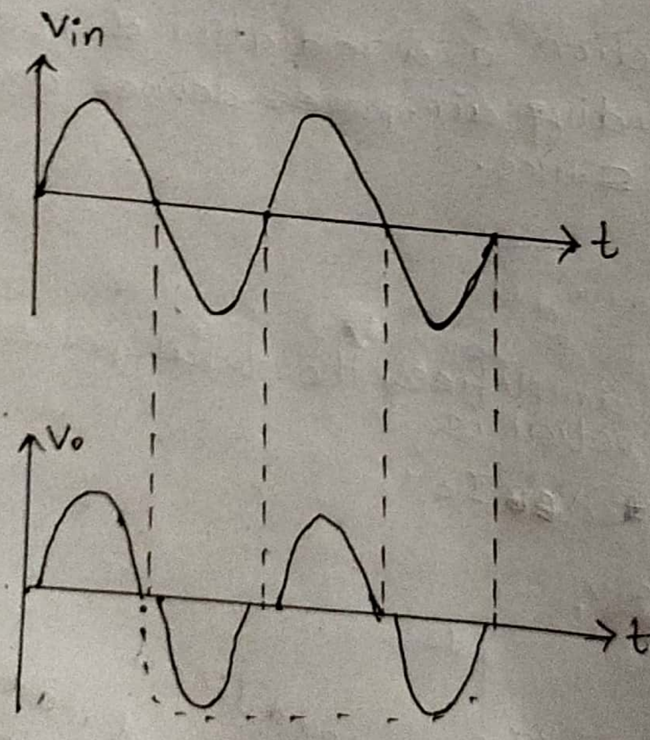


fig. Cross overdistortion

5.b. A negative feedback of 0.2 % is applied to an amplifier with gain of 60 dB. calculate the percentage change in the overall voltage gain of the feedback amplifier if the internal amplifier is subjected to gain of reduction of 15%.

Sol Given:

Feedback ratio, $\beta = 0.2\% = 0.002$

open loop gain, $A_{dB} = 60$ dB

change in open loop gain, $\frac{dA}{A} = 15\%$

change in feed back gain, $\frac{dA_f}{A_f} = ?$

we know

$$A_{dB} = 20 \log A$$

$$\Rightarrow 60 = 20 \log A$$

$$\Rightarrow \frac{60}{20} = \log_{10} A$$

$$\Rightarrow A = 10^3 = 1000$$

Now,

$$\frac{dA_f}{A_f} = \frac{1}{(1+\beta A)} \times \frac{dA}{A}$$

$$= \frac{1}{(1+0.002 \times 1000)} \times 15\%$$

$$= \frac{15}{3}\%$$

$$\therefore \frac{dA_f}{A_f} = 5\%$$

Thus the percentage change in the overall voltage gain is 5%. A

5.b. What are positive and negative feedback? Negative feedback extends the bandwidth. Explain it with necessary derivations.

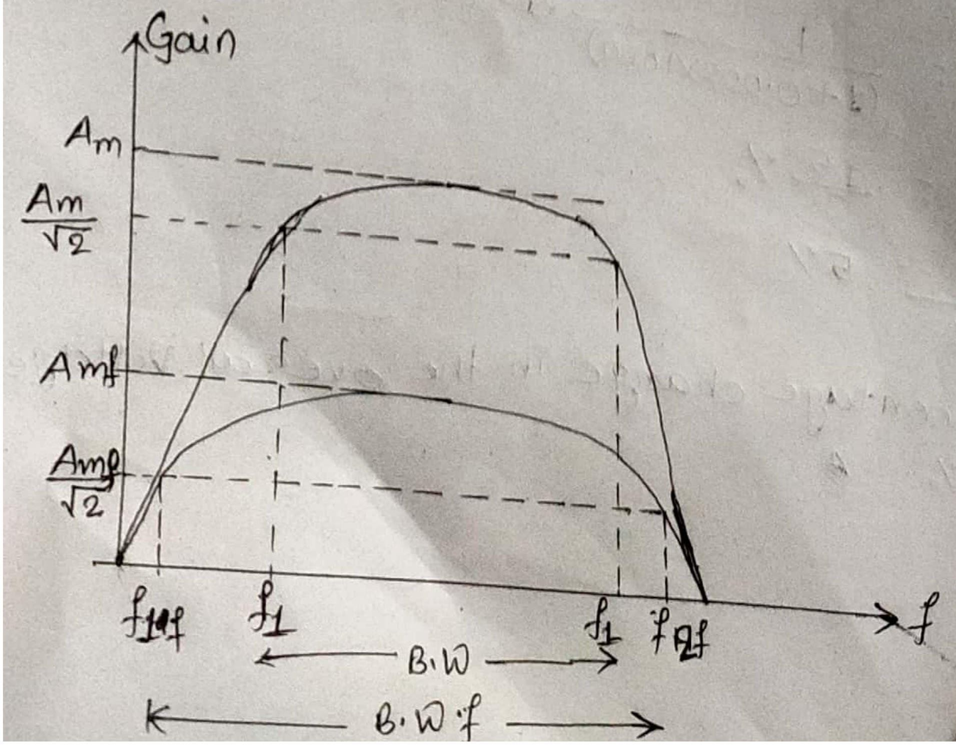
Sol^n: Positive feedback is the feedback which is used to increase the input signal. It is also called direct or regenerative feedback.

Negative feedback is the feedback which is applied to reduce the input signal. It is also called inverse or degenerative feedback.

Negative feedback extends the bandwidth:

If f_L and f_H are lower and upper cut off frequency of an amplifier without feedback then the values of f_L decreases and f_H increases by factor of $(1+A\beta)$, if negative feedback is used.

The plot of gain and frequency of an amplifier is shown below:



Here, without feedback,

$$BW = f_2 - f_1$$

where f_1 = lower cut off frequency

f_2 = higher/upper cut off frequency

After negative feedback,

$$BW_f = f_{2f} - f_{1f}$$

$$\Rightarrow BW_f = f_2(1+\beta A_m) - \frac{f_1}{(1+\beta A_m)}$$

$$\Rightarrow BW_f = f_2(1+\beta A_m) \quad (\because \gg \frac{f_1}{(1+\beta A_m)})$$

$$\Rightarrow BW_f = BW * (1+\beta A_m) \; (\because BW = f_2 - f_1 \approx f_2)$$

$$\therefore BW_f = BW(1+\beta A)$$

Thus the negative feedback extends bandwidth